

Real Numbers

Euclid's Division Lemma, the Fundamental Theorem of Arithmetic, and proving that a number is irrational, with worked examples and practice questions.

Introduction

Real numbers are all the numbers that can be placed on the number line. They include the rational numbers (which can be written as p/q , where p and q are integers and $q \neq 0$) and the irrational numbers (which cannot). In this chapter, we look closely at positive integers through two results: **Euclid's Division Lemma** and the **Fundamental Theorem of Arithmetic**. We then use them to decide whether certain numbers are rational or irrational.

Euclid's Division Lemma

This result is really just a formal statement of the long division you have done since primary school: when you divide one number by another, you get a quotient and a remainder, and the remainder is always smaller than the number you divided by.

EUCLID'S DIVISION LEMMA

Given positive integers a and b , there exist unique integers q and r such that

$$a = bq + r, \text{ where } 0 \leq r < b$$

Here, a is the dividend, b is the divisor, q is the quotient and r is the remainder.

The word "lemma" means a proven statement used as a stepping stone to prove a bigger result. Here, it leads us directly to a simple, reliable way of finding the HCF of two positive integers.

Euclid's Division Algorithm

Euclid's Division Algorithm is a repeated application of the division lemma, used to find the **HCF (Highest Common Factor)** of two positive integers.

To find the HCF of two positive integers, say c and d , with $c > d$, follow these steps:

Step 1. Apply Euclid's Division Lemma to c and d . This gives whole numbers q and r such that $c = dq + r$, where $0 \leq r < d$.

Step 2. If $r = 0$, then d is the HCF of c and d . If $r \neq 0$, apply the division lemma again, this time to d and r .

Step 3. Continue this process until the remainder becomes 0. The divisor at that final step is the HCF of c and d .

WORKED EXAMPLE

Use Euclid's Division Algorithm to find the HCF of 455 and 42.

Since $455 > 42$, we start by dividing 455 by 42:

$$455 = 42 \times 10 + 35$$

The remainder 35 is not 0, so we now divide 42 by 35:

$$42 = 35 \times 1 + 7$$

The remainder 7 is not 0, so we divide 35 by 7:

$$35 = 7 \times 5 + 0$$

The remainder is now 0, and the divisor at this step is 7. So, **HCF(455, 42) = 7**.

Euclid's Division Algorithm works because the HCF of two numbers does not change if we replace the larger number by its remainder when divided by the smaller one. This is exactly what each step of the algorithm does.

The Fundamental Theorem of Arithmetic

You already know that any composite number can be broken down into a product of primes (for example, $30 = 2 \times 3 \times 5$). This chapter states that fact formally, and adds an important detail: this factorisation is *unique*.

FUNDAMENTAL THEOREM OF ARITHMETIC

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

WORKED EXAMPLE

Express 140 as a product of its prime factors.

$$140 = 2 \times 70 = 2 \times 2 \times 35 = 2 \times 2 \times 5 \times 7 = 2^2 \times 5 \times 7$$

No matter which pair of factors you start with, you will always arrive at the same set of prime factors. This is what "unique factorisation" means.

HCF and LCM by Prime Factorisation

The Fundamental Theorem of Arithmetic gives us a second method for finding the HCF and LCM of two or more positive integers, often quicker than Euclid's Algorithm when the numbers factorise easily.

HCF = product of the smallest power of each common prime factor of the numbers.

LCM = product of the greatest power of each prime factor, of all the numbers, involved.

WORKED EXAMPLE

Find the HCF and LCM of 6 and 20 by prime factorisation.

$$6 = 2 \times 3 \quad 20 = 2^2 \times 5$$

The only common prime factor is 2, and its smallest power here is 2^1 , so $\text{HCF}(6, 20) = 2$.

For the LCM, take the greatest power of every prime factor that appears: $2^2 \times 3 \times 5 = \mathbf{60}$.

For any two positive integers a and b , it is always true that:

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

You can check this in the example above: $\text{HCF} \times \text{LCM} = 2 \times 60 = 120$, and $a \times b = 6 \times 20 = 120$. This relationship is a useful check on your working, and is sometimes used directly in exam questions to find an LCM once the HCF is known (or vice versa), but note that it only holds for exactly two numbers, not three or more.

Proving a Number is Irrational

A number is **irrational** if it cannot be written in the form p/q , where p and q are integers and $q \neq 0$. In this section, we prove that certain square roots are irrational. This is one of the most important proof-based questions in this chapter, and it appears frequently in board exams.

THEOREM

$\sqrt{2}$ is irrational.

Proof. We prove this by contradiction: we assume the opposite of what we want to prove, and show that this assumption leads to something impossible.

Let us assume, to the contrary, that $\sqrt{2}$ is rational. Then we can write

$$\sqrt{2} = p/q$$

where p and q are coprime integers (that is, $\text{HCF}(p, q) = 1$) and $q \neq 0$. This is always possible, since any rational number can be reduced to its simplest form.

Squaring both sides:

$$2 = p^2/q^2 \Rightarrow p^2 = 2q^2 \dots(1)$$

From (1), p^2 is divisible by 2. It follows that p itself must be divisible by 2: if a prime number divides the square of an integer, it must also divide the integer itself, by the Fundamental Theorem of Arithmetic. So we can write $p = 2c$ for some integer c .

Substituting $p = 2c$ back into (1):

$$(2c)^2 = 2q^2 \Rightarrow 4c^2 = 2q^2 \Rightarrow q^2 = 2c^2$$

This shows q^2 is also divisible by 2, and so q must be divisible by 2 as well.

But this means both p and q are divisible by 2, which contradicts our assumption that p and q are coprime ($\text{HCF}(p, q) = 1$). This contradiction has arisen because of our incorrect assumption that $\sqrt{2}$ is rational.

Hence, **$\sqrt{2}$ is irrational.**

The same method (assume the number is rational, write it in the form p/q with p and q coprime, and show that this forces both p and q to share a common factor) can be used to prove that $\sqrt{3}$, $\sqrt{5}$, and in general $\sqrt{p}$ for any prime number p , are irrational.

WORKED EXAMPLE

Prove that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational.

Assume, to the contrary, that $3 + 2\sqrt{5}$ is rational. Then we can write

$$3 + 2\sqrt{5} = p/q$$

for some coprime integers p and q , with $q \neq 0$. Rearranging,

$$2\sqrt{5} = p/q - 3 = (p - 3q)/q \Rightarrow \sqrt{5} = (p - 3q)/2q$$

Since p and q are integers, the right-hand side $(p - 3q)/2q$ is a rational number. This would mean $\sqrt{5}$ is rational, but this contradicts the known fact that $\sqrt{5}$ is irrational.

This contradiction shows our assumption was wrong. Hence, **$3 + 2\sqrt{5}$ is irrational.**

This same reasoning applies generally: the sum of a rational number and an irrational number is always irrational, and the product of a nonzero rational number and an irrational number is always irrational.

Summary

- **Euclid's Division Lemma:** for positive integers a and b , $a = bq + r$, where $0 \leq r < b$.
- **Euclid's Division Algorithm** uses the lemma repeatedly to find the HCF of two positive integers.

- **Fundamental Theorem of Arithmetic:** every composite number can be factorised into primes in exactly one way, apart from the order of the factors.
- HCF = product of the smallest power of each common prime factor; LCM = product of the greatest power of each prime factor involved.
- For any two positive integers, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.
- $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, and $\sqrt{p}$ for any prime p , are irrational (proved by contradiction).
- A rational number plus (or times) an irrational number is always irrational.

Practice Questions

1. Use Euclid's Division Algorithm to find the HCF of 196 and 38220.

2. Express 156 as a product of its prime factors.

3. Find the HCF and LCM of 26 and 91, and verify that $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$.

4. Prove that $\sqrt{3}$ is irrational.

5. Prove that $5 - \sqrt{2}$ is irrational.

6. Two tankers contain 850 litres and 680 litres of fuel. Find the maximum capacity of a container that can measure the fuel of both tankers an exact number of times.